Vortical Enclosure Dynamics, Vol. 4

The Differential Horizon: Divergence, Renormalization, and the Boundary of Generation

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Abstract

We propose the *Differential Horizon* as a structural framework for reinterpreting divergence, renormalization, and the limits of physical description. Rather than treating divergence only as a technical failure to be corrected, we read it as a possible boundary signal: an indication that a descriptive window has been pushed toward the limit of the differences it can represent as finite quantities.

From this perspective, renormalization is not an operation that erases a boundary. It is a finite re-expression of boundary contact inside a chosen observational window. The speed-of-light limit, the Planck-scale breakdown of spacetime continuity, and black-hole singularities are re-positioned as local or hierarchical horizon-forms, rather than as unrelated failures or as different objects beyond theory.

This framework does not propose a unified theory of interactions. Instead, it identifies a prior structural layer: the condition under which registered difference, trace formation, and physical description become possible. The *Differential Horizon Principle* states that a description based on difference cannot convert the absence of difference into an ordinary observable state.

VED (Vortical Enclosure Dynamics) is positioned within this framework as an open generative description. It does not explain existence from outside existence. It describes why generated spacetime encounters a descriptive boundary when it attempts to reach the condition from which its own differences arise.

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1 Introduction

The problem of renormalization has stood at the center of modern physics for decades. Quantum field theory produces divergent quantities during calculation; renormalization absorbs these into observable definitions and yields predictions of extraordinary precision. Yet the physical meaning of this procedure remains difficult to state.

This volume takes that difficulty as a structural clue. We do not treat divergence merely as a mathematical artifact to be removed. We read it as a possible boundary signal: the trace of a descriptive window being pushed toward the limit of the differences it can represent as finite quantities.

The name given here to this limit is the *Differential Horizon*. It is not a wall located somewhere in spacetime, nor a hidden object beyond physical theory. It is a descriptive boundary that appears when a description based on registered difference attempts to describe the absence of difference as if it were one more observable state.

Three claims organize this volume. First, physical description operates on generated traces: differences that have become stable enough to be registered, compared, and related. Second, divergence is not merely failure, but may signal boundary contact with the limit of a given observational window. Third, renormalization is not simple elimination, but finite re-expression: it reorganizes variables, scales, and reference structures so that finite relations can still be extracted.

From this perspective, the speed-of-light limit, the Planck-scale breakdown of spacetime continuity, black-hole singularities, and quantum-field divergences are not treated as unrelated failures. Nor are they collapsed into one literal hidden object. They are read as local or hierarchical horizon-forms: limits that recur when a successful descriptive framework is pushed toward the disappearance, compression, or overclosure of the differences on which it depends.

This framework does not propose a final unified theory. It identifies a prior structural layer: the condition under which registered difference, trace formation, and physical explanation become possible. That condition is made explicit in the *Differential Horizon Principle*.

VED is positioned within this framework as an open generative description. It does not explain existence from outside existence. It describes why generated spacetime encounters a descriptive boundary when it attempts to erase the differences that make its own description possible.

The paper proceeds as follows. Section 2 examines the unease of renormalization and reframes it as a structural clue. Sections 3–4 establish that physics operates on pastified traces and that the *un-generated* cannot be converted into an object of description. Section 5 introduces the *Differential Horizon* at an intuitive level.

Sections 6–8 develop the reinterpretation of divergence and renormalization. Sections 9–10 re-place major theories and unsolved problems within the framework. Section 11 states the Differential Horizon Principle. Section 12 positions VED. Section 13 gives the smallest dynamical form of this principle as a local *Differential Horizon*. Section 14 concludes.

2 The Unease of Renormalization

Renormalization is one of the most successful procedures in modern physics. Quantum field theory produces divergent quantities during calculation; renormalization absorbs these divergences into observable definitions and yields predictions that agree with experiment to extraordinary precision.

Yet this success conceals a fundamental unease.

The procedure works with remarkable accuracy, yet its conceptual meaning remains unresolved. A theory that produces infinities mid-calculation is nevertheless able to describe physical reality correctly after appropriate redefinition.

This produces a peculiar situation.

A theory breaks down once, yet after modification, it succeeds.

Why does a theory that diverges still succeed?

This question is rarely pursued at a foundational level. Renormalization has been formalized and refined, and its mathematical structure is well understood. However, its physical meaning remains unclear.

What, precisely, is renormalization doing?

Why do divergences appear? And why does removing them yield correct predictions?

No unified interpretation of these questions has been established.

In this paper, we take this unease seriously.

Rather than treating renormalization as a problem to be resolved, we treat it as a structural clue.

The appearance of divergence may not indicate a failure of theory, but contact with the limit of its descriptive window.

From this perspective, renormalization is not merely a technical operation. It may be a procedure that converts boundary contact into finite relations that can be handled within the theory.

This reinterpretation will be developed throughout this volume.

Divergence is not removed; it is transformed.

And what appears as infinity may be the signal of a descriptive boundary.

3 Physical Description as Pastified Texture

The preceding discussion suggested that divergence may not be a mere failure of theory, but a signal that theory is approaching a boundary of its own conditions. To understand this possibility, we must first examine what physical description actually describes.

Physics does not directly describe the living instant of generation. It describes differences that have already become stable enough to be registered, compared, and related. Particles, fields, energies, events, and coordinates do not describe raw pre-generative flow directly. They are generated residues that have acquired enough stability to enter description.

> Physics describes registered difference.

This does not mean that physics is false, subjective, or merely retrospective in a psychological sense. It means that physical description requires trace formation. A trace is already a partial past: a residue of motion that can be compared with other residues, measured against a standard, or placed into a relation. Without such a trace, there is no object of measurement.

The past here does not simply mean what has elapsed in time. It means what has become available as a distinguishable form. Even when physics predicts the future, it begins from registered conditions: states, fields, parameters, or probability assignments that have already been stabilized enough to enter calculation. Prediction extends a texture of registered differences; it does not directly grasp the pre-generative source from which difference first becomes possible.

Spacetime as Organized Texture

Spacetime should therefore not be treated as the primitive container inside which traces are simply placed. Rather, spacetime is the organized texture through which generated traces become mutually relatable. It supplies the relational order by which events can be compared, distances can be assigned, durations can be measured, and motions can be described.

In this sense, spacetime is already part of the generated order. It is not outside the texture of description. It is the texture in which generated residues become comparable.

> Physical description operates on generated residues,
> not on the pre-generative source itself.

This gives a precise meaning to *pastified texture*. Pastification is the stabilization of difference into traces that can be handled by description. Texture is the relational organization of those traces into a form that can be measured, compressed, renormalized, or compared across scales.

Color, temperature, mass, field value, and curvature are not merely isolated qualities. They are stabilized textures of relation. They arise when underlying differences have become sufficiently organized to be treated as describable quantities.

The Limit of Trace Formation

This structure also clarifies why a boundary appears. If physical description operates on registered differences, then it cannot directly convert the absence of difference into an observable state. Where no difference is available, there is no trace. Where no trace can form, there is nothing to compare, measure, or relate.

The *un-generated* must therefore not be understood as a hidden object behind physical description. It is not a concealed region waiting to be observed. It names the limit at which the formation of describable traces is no longer available.

> The absence of difference cannot become one more registered difference.

From this perspective, divergence is not simply a computational defect. It marks the point at which a framework that operates on pastified texture is pushed toward the condition where texture itself can no longer be constituted. The equation does not merely fail to produce a number. It indicates that the observational and mathematical window has reached the edge of the differences it can stabilize.

Bridge to the Differential Horizon

The next sections develop this boundary more explicitly. Section 5 first introduces the *Differential Horizon* at an intuitive level: if description operates on registered differences, then it cannot directly convert the absence of difference into an observable state. Section 11 later formulates this structure as the *Differential Horizon Principle*.

> Physics holds within the world of pastified texture.
> Of the condition before trace formation, physics can speak only through its boundary.

4 Why the Un-generated Cannot Be Handled

The previous section established that physical theory operates on pastified texture: registered differences that have become stable enough to be compared, measured, and related.

This naturally raises a question.

> Why can the *un-generated* not be handled as an object of description?

One might initially assume that the *un-generated* is simply unknown, and that a more advanced theory will eventually represent it.

However, the issue here is not merely one of incomplete knowledge.

It concerns the form of description itself.

Physical theory presupposes several basic structures: time, space, states, observable quantities, and registered difference. These are taken as given, and laws are defined on the basis of them.

The *un-generated* fits none of these presuppositions.

If time has not yet come to hold, the question of "when" has no meaning. If space has not yet come to hold, the question of "where" does not arise. If states are not defined, quantities cannot be assigned. If observable quantities do not exist, theory has no object. If difference has not become registerable, there is no trace on which observation can operate.

From this, a crucial point follows.

> The *un-generated* is not merely unknown.
> It cannot be converted into an ordinary object of description.

It does not fit the form of description itself, because description already presupposes the differences whose absence is at issue.

No matter how precise a theory becomes, as long as it presupposes structures such as time and space, it cannot turn the absence of those structures into one more state inside them.

This is not a limitation of current theory, but a constraint imposed by the conditions under which theory holds.

> Because theory presupposes registered difference,
> it cannot describe the absence of difference as a difference.

At this point, the significance of divergence becomes clearer.

Theory cannot describe the *un-generated* directly. But a theory can be pushed toward the limit at which its own variables cease to stabilize finite differences.

As it does so, quantities cease to be defined, and divergence appears.

This is not accidental.

It indicates that the descriptive window has reached a limit: the point at which its variables can no longer preserve the differences they were designed to represent.

However, this boundary is not a sharp line.

> It appears as a loss of descriptive stability.

Physical theories always operate within finite precision and scale. Their observational windows do not fail all at once. They lose stability gradually, as their variables are compressed, amplified, or iterated beyond the range in which they remain finite.

Within this loss of stability, theory does not immediately fail, but becomes unstable, and eventually produces divergence.

From this perspective, the following statement becomes natural.

> Divergence is not access to the *un-generated*.
> It is the trace of a description reaching its own limit.

This contact does not allow direct description. But it cannot be denied.

Theory approaches, responds, and then breaks down. The manner of this breakdown itself is the only available indication of the boundary.

Thus, the *un-generated* does not appear within theory directly. It appears only as the failure of a generated description to convert absence of difference into an observable state.

The next section defines this boundary explicitly.

It is not introduced as an abstract limit, but as a structure with physical meaning: the *Differential Horizon*.

5 The Differential Horizon

From the preceding discussion, one necessity has become clear. Physical theory does not observe an object-in-itself directly. Observation registers difference within a generated domain.

This point is elementary, but it changes the meaning of the boundary considered here. If VED begins from difference, then observation can operate only where difference has already become describable. An observation window can register contrast, variation, gradient, motion, or event. It cannot register the absence of difference as though that absence were one more ordinary state.

> Observation begins where difference is already available.

If no difference is available, there is no gradient. If there is no gradient, there is no motion. If there is no motion, there is no distinguishable event for observation to register. The absence of difference therefore cannot be made into one more difference inside the same generated domain.

This is the first intuitive appearance of the *Differential Horizon*. It is not yet the formal *Differential Horizon Principle*. It is the descriptive boundary that appears when a description based on difference tries to convert the absence of difference into an ordinary object of description.

> The *Differential Horizon* is the limit at which difference-based description fails to convert the absence of difference into an observable state.

The *Differential Horizon* should therefore not be understood as a hidden place or a literal spatial surface. It is a boundary of describability. It marks the point at which the observational conditions of a generated domain no longer suffice to describe what would erase those conditions.

In this sense, the *Differential Horizon* is not a concept for describing an external object. It is a concept for describing why such an objectification fails. The *un-generated* must not be treated as a concealed region waiting to be observed. It appears only as the limit of an observation window that depends on difference.

> The absence of difference cannot be observed as a difference.

From this perspective, divergence acquires its first interpretation. Equations do not fail arbitrarily when they approach such a boundary. They signal that a description has reached the edge of the differences it can resolve. Later sections will refine this claim through spatial convergence, temporal convergence, renormalization, and the relation between multiple observational windows reaching the same descriptive limit.

Figure 1 illustrates the overall structure: the *un-generated* limit, the *Differential Horizon* as a descriptive boundary, and the *generated* domain in which texture, symmetry, and conserved quantities emerge through compression and coarse-graining.

The full structural formulation will be given in Section 11, where this intuitive boundary is stated explicitly as the *Differential Horizon Principle*. The next section first examines how approach to this boundary appears in concrete terms, through two distinct modes: spatial convergence and temporal convergence.

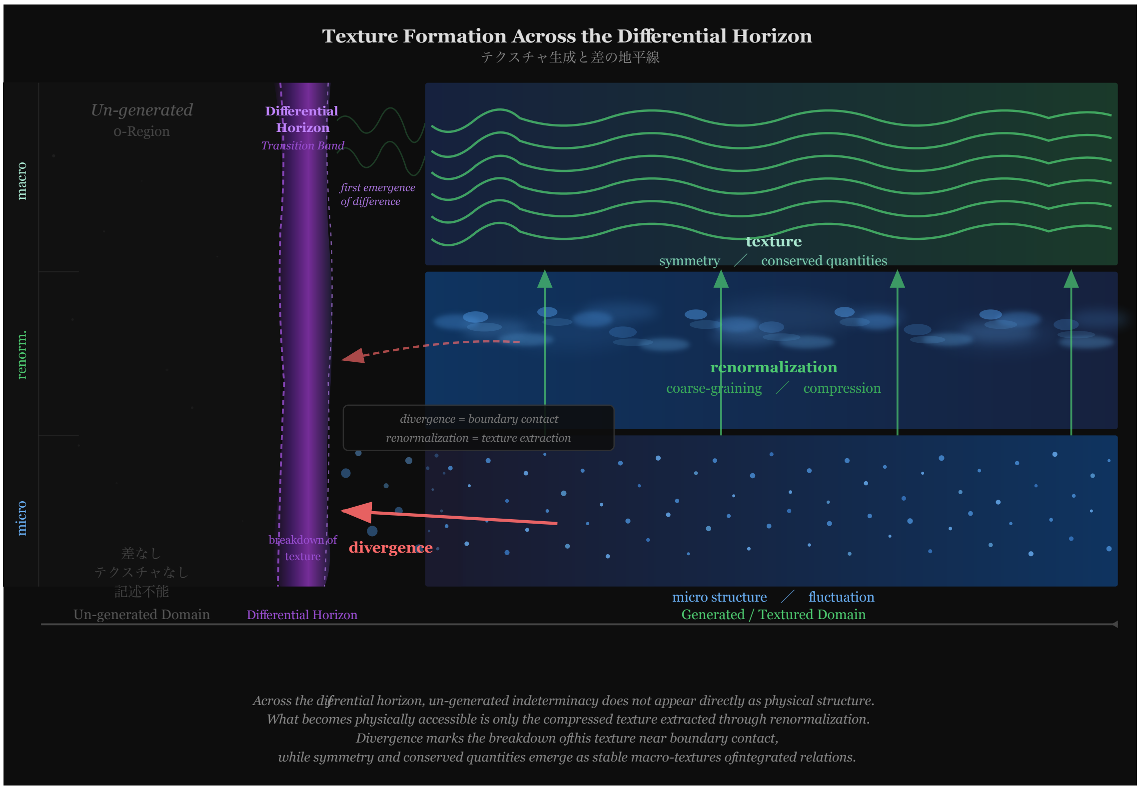

Figure 1: Texture formation across the *Differential Horizon*. The *un-generated* limit is not an observable object or region; it marks the absence of describable difference. The transition band represents the *Differential Horizon* as a descriptive boundary. The *generated* domain is organized into three layers: micro-structure and fluctuation (bottom), renormalization and coarse-graining (middle), and stable texture, symmetry, and conserved quantities (top). Divergence marks the breakdown of texture near *boundary contact. The world appears not as raw microstructure, but as compressed texture.*

6 Two Paths of Approach

The *Differential Horizon* cannot be reached as an object. Its effects appear when a descriptive window is pushed toward its limit.

Within generated spacetime, this occurs through different observational windows.

> Spatial convergence and temporal convergence are dual ways in which descriptive limits become visible.

6.1 Spatial Convergence

Approaching ever smaller scales leads into the quantum domain.

Particles lose classical localization, fields fluctuate strongly, and quantities such as position and momentum cease to be simultaneously definable.

As the scale decreases further, theory gradually loses stability.

Quantum fluctuations intensify, and calculations develop divergences.

At the Planck scale, the continuity of spacetime itself is expected to break down.

Distance and position lose meaning.

> Reducing scale exposes a local horizon-form.

This path is described by quantum theory and quantum field theory. Divergence appears frequently, and renormalization becomes indispensable.

6.2 Temporal Convergence

The second path arises through increasing velocity.

In relativity, as velocity approaches the speed of light, time dilation becomes extreme.

At the limit, the required energy for a massive body diverges, and the temporal structure of the frame is pushed to its boundary.

Simultaneously, length contraction eliminates spatial extension along the direction of motion.

> Approaching the speed of light exposes a temporal horizon-form.

This path is described by relativity. Energy diverges toward infinity, and a body with finite mass cannot reach the speed of light. The structure of theory itself does not permit it.

6.3 Unified Structure

Though these two paths appear fundamentally different, they share a common structural feature.

> Both paths lead to limits in the definability of spacetime structure.

In spatial convergence, fluctuation destroys localization.
In temporal convergence, extreme motion destroys temporal duration.
In both cases, difference itself ceases to be well-defined.

> Different paths.
> Related losses of definability.

This is not coincidence.

The problem of quantum gravity, viewed from this perspective, is not simply a matter of integrating two theories. It is a matter of understanding the structure by which different observational windows expose related limits of spacetime description.

> They are not different domains, but different trajectories through related horizon-forms.

These approaches do not reveal an object beyond the boundary. They reveal the recurrent condition under which the structures that make physical description possible begin to lose stability.

This structure is illustrated in Figure 2.

The next section redefines divergence formally in light of this convergent structure.

7 Divergence as Boundary Signal

From the preceding discussion, one conclusion becomes unavoidable. Physics operates on registered differences. Its equations describe quantities that have become stable enough to be compared, measured, or related inside a generated domain.

A divergence should therefore not be interpreted too quickly as the literal appearance of an actual infinite object. It may indicate that the variables of a theory are being asked to represent differences beyond the scale, texture, or reference structure in which those variables remain finite.

> Divergence is not simply an infinite object appearing inside theory.

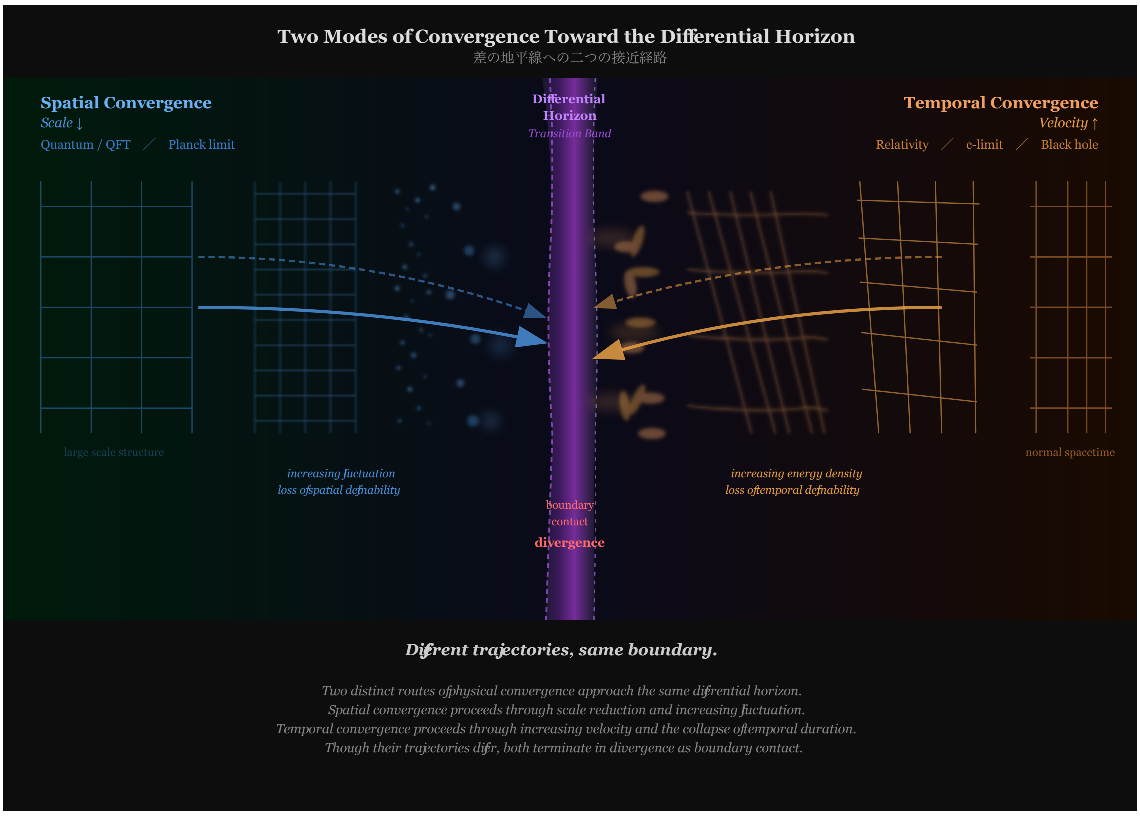

Figure 2: Two modes of convergence toward the *Differential Horizon*. Spatial convergence (left, blue) proceeds through scale reduction and increasing fluctuation toward the Planck limit. Temporal convergence (right, orange) proceeds through velocity increase and the collapse of temporal duration toward the speed-of-light limit. Both routes terminate in divergence as *boundary contact*. *Different trajectories, related descriptive limits.*

Nor is divergence merely a failure of rigor. In many cases, a divergence signals that a descriptive window has been pushed beyond the differences it can stably register. The theory remains meaningful within its domain, but its variables may no longer be able to represent the required differences as finite quantities.

Registered Difference and Overextension

The point is structural. Physical description depends on traces, scales, and quantities that have already become describable. When a theory compresses difference toward a point, amplifies difference toward unbounded energy, or iterates difference beyond the stability of its own variables, divergence can appear.

From the perspective of VED, divergence is read as a boundary signal: the trace of a description being pushed beyond the differences it can stably register.

> Divergence marks descriptive overextension.

This statement should not be overread. VED does not claim that all divergences have the same physical origin. Different theories, scales, and mathematical structures produce divergences in different ways. The claim is interpretive and structural: divergences can indicate that a successful descriptive window has approached a local boundary of finite representation.

What Divergence Does Not Reveal

Divergence does not turn the boundary into an object of description. It does not provide access to a hidden region, nor does it turn the *un-generated* into something observable. It reflects the limit of the current description.

> Divergence does not cross the boundary.
> It marks the limit of a window.

This is why divergence cannot be treated only as a value to be discarded. It may also be a diagnostic sign. It indicates that the theory has reached the edge of the differences its own variables can carry as finite quantities.

Implication for Renormalization

At this point, the meaning of renormalization begins to shift. If divergence marks boundary contact or descriptive overextension, then renormalization is not simply the deletion of a mistake. It is the finite re-expression of that contact inside a chosen observational or mathematical window.

Divergence signals boundary contact;
renormalization re-expresses that contact finitely.

This will be examined in detail in the next section.

8 Reinterpreting Renormalization

The previous section redefined divergence as a signal that a descriptive window has reached the edge of the differences it can represent. From this perspective, renormalization also changes its meaning. It is not merely a technical device for removing infinities. It is a finite re-expression of boundary contact.

Renormalization does not erase the boundary.
It converts contact with the boundary into finite relations.

A divergence is not merely a mathematical defect. It can indicate that the variables, scales, or reference structure of the current description can no longer represent the required differences as finite quantities. The theory has not thereby become useless. It has reached the limit of a particular observational window.

Finite Re-Expression

Renormalization responds to this limit by reorganizing the description. It changes which variables are treated as primitive, which scales are resolved, and which quantities are taken as reference values. Through this reorganization, finite relational quantities can still be extracted.

Renormalization is not elimination.
It is controlled re-expression.

This re-expression does not turn the *Differential Horizon* into an object of description. It does not cross the boundary or make the *un-generated* observable. It changes the scale and variables through which the boundary is represented from within a generated domain.

In the language developed here, renormalization is a controlled response to a local *Differential Horizon*. The local horizon is not a new object. It is the limit of a descriptive window: the point where that window can no longer represent the required differences without reorganization.

Effective Description

Effective theories make this structure explicit. An effective theory remains valid not because it has eliminated the deeper boundary, but because it has stabilized a finite texture of relations inside a limited window. It selects the differences that remain resolvable at that scale, compresses unresolved structure into parameters, and preserves the relations needed for prediction.

> An effective description is finite because it is windowed.

This is why scale dependence is not an accidental complication. Different scales expose different limits of representation. At one scale, a quantity can be treated as finite and stable. At another scale, the same description may encounter a divergence and require re-expression.

From the perspective of VED, this is not a replacement for renormalization theory. It is a generative reading of why renormalization is needed: description survives boundary contact by changing the texture through which finite relations are expressed.

Boundary Contact and Later Horizons

Divergence and renormalization are therefore not opposing operations. Divergence signals the edge of a descriptive window. Renormalization reorganizes that window so that finite relations can still be extracted.

> Divergence marks local boundary contact;
> renormalization makes finite description possible after that contact.

This interpretation prepares the later discussion of unresolved problems. Quantum field theory divergences, Planck-scale breakdown, black-hole singularities, and other limits will be read not as isolated failures, nor as one hidden object behind physics, but as local or hierarchical horizon-forms of description.

The next section places major physical theories within this framework, before Section 10 returns to the horizon-like limits of modern physics.

9 Theoretical Re-placement

From the preceding discussion, the structure of the *Differential Horizon* and the meaning of divergence and renormalization have become clear.

From this perspective, modern physical theories can be reorganized.

> Theories are not classified by their objects,
> but by the observational windows they stabilize.
>
> Each window has its own local horizon-form.

9.1 Quantum Field Theory

Quantum field theory stabilizes a field-theoretic window of registered difference.

At high energies, field fluctuations increase, and calculations exhibit divergence.

These divergences are handled through renormalization, allowing finite predictions.

> Quantum field theory remains effective by re-expressing boundary contact as finite quantities.

9.2 Relativity

Relativity stabilizes the spacetime window in which motion, duration, and causal order are mutually defined.

As velocity approaches the speed of light, time dilation becomes extreme.

Under strong gravitational fields, spacetime structure breaks down, and singularities appear.

> Relativity exposes horizon-like limits when temporal or gravitational closure exceeds the descriptive stability of spacetime variables.

9.3 Black Holes

Black holes represent one of the most explicit gravitational horizon-forms.

At the event horizon, the structure of spacetime becomes inaccessible to external observers.

At the singularity, curvature diverges, and spacetime loses definability.

> Black holes show how gravitational closure can push a descriptive frame beyond its ability to preserve distinguishable structure.

9.4 String Theory

String theory emerges as an attempt to maintain coherence where point-field descriptions and gravitational descriptions both approach their limits.

Its mathematical structure seeks to suppress divergence and maintain internal consistency.

> String theory explores whether a different descriptive window
> can preserve finite relations near these limits.

9.5 Quantum Gravity

Quantum gravity is often described as the problem of unifying quantum theory and relativity.

From the perspective of the *Differential Horizon*, its meaning shifts.

> Quantum gravity is not a missing theory,
> but a structural problem of convergence.

It concerns how different descriptive windows fail, overlap, or need to be re-expressed when spacetime itself can no longer be treated as a stable background texture.

9.6 Unified Structure

These theories are not independent, but neither do they reduce to one hidden object behind them.

> These theories do not contradict one another.
> They stabilize different windows within a related structure.

Quantum field theory approaches through scale. Relativity approaches through velocity and curvature.

Black holes expose gravitational overclosure. String theory explores coherence near the loss of point-like description.

Quantum gravity expresses the difficulty of understanding how these windows meet when their variables cease to remain independently stable.

> All theories are finite organizations of registered difference.

Figure 3 maps these positions.

The next section shows how this unified view gathers the major unsolved problems of modern physics as local or hierarchical horizon-forms.

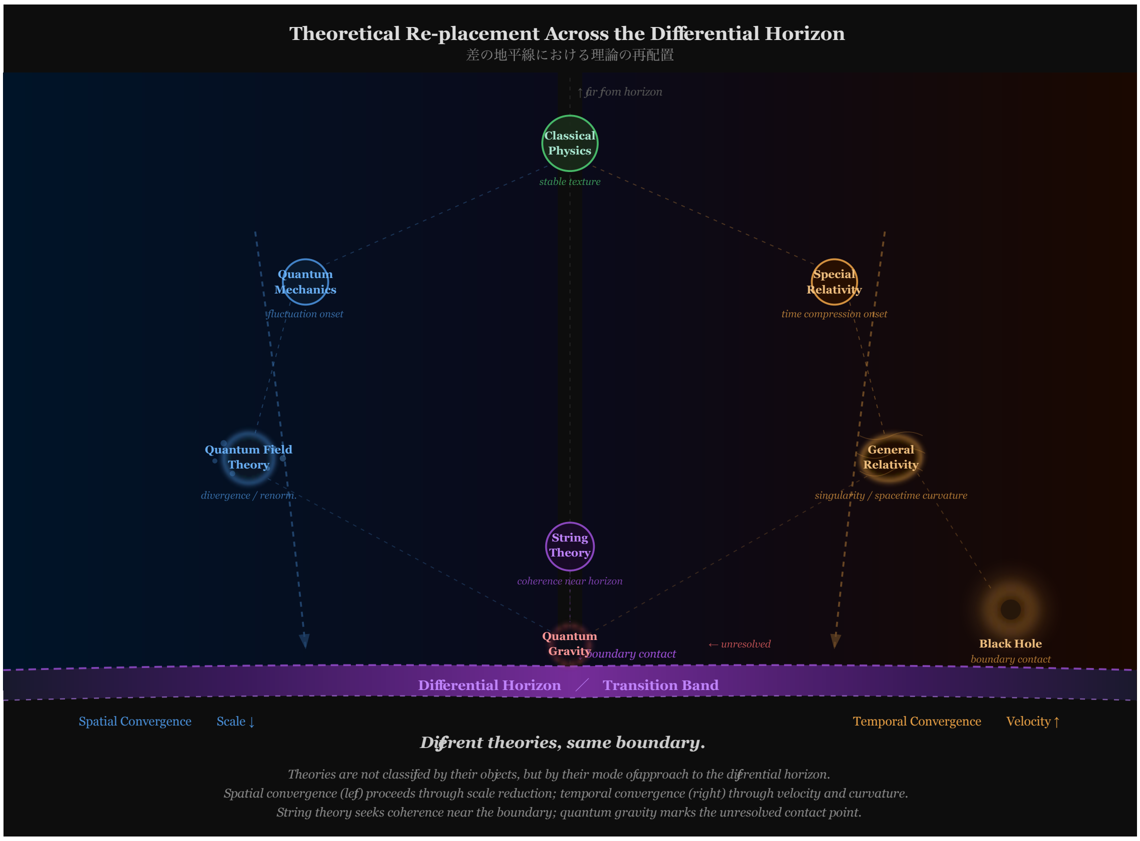

Figure 3: Theoretical re-placement across local horizon-forms. Theories are organized not by their objects but by the observational windows they stabilize. Spatial convergence (left) encompasses quantum mechanics and quantum field theory. Temporal convergence (right) encompasses special and general relativity. Black holes occupy the *boundary contact* region. String theory seeks coherence where point-like description loses stability. Quantum gravity marks the unresolved contact point. The phrase "same boundary" in the schematic should be read structurally: different theories expose related local horizon-forms within their own descriptive windows. *Different windows, related descriptive limits.*

10 Unresolved Problems as Local Horizons

Modern physics contains many unresolved problems. Quantum field theory divergences, Planck-scale breakdown, black-hole singularities, and the speed-of-light limit are usually treated as belonging to different domains. They should not be collapsed into a single hidden object, nor should they be treated as unrelated failures.

From the present perspective, they are horizon-like limits that arise when a successful descriptive window is pushed toward the condition its own variables cannot represent. Each window has its own variables, scales, and stabilizing assumptions. When those assumptions are driven toward disappearance, compression, or overclosure of difference, a local *Differential Horizon* appears.

> A local *Differential Horizon* appears when a theoretical or observational window reaches the limit of the differences it can resolve.

This does not mean that there are many independent ultimate horizons. It means that the same structural form can appear hierarchically in different domains. Here, "hierarchical" does not mean a stack of hidden regions. It means that horizon-like limits recur when the observational window is rescaled or reorganized. Different theories generate horizon-like limits with similar structure but different physical expressions.

10.1 Divergence in Quantum Field Theory

Quantum field theory is extraordinarily successful within its domain. Its divergences should therefore not be read as simple failure. From the present perspective, they appear when field-theoretic resolution is pushed toward scales where the assumed texture of fields, points, and local interactions can no longer represent the required differences as finite quantities.

Renormalization does not make this limit disappear. It reorganizes the divergent structure into finite observable quantities within a coarser descriptive window. In this sense, renormalization is a way of preserving description after a local horizon has been encountered.

10.2 Planck-Scale Breakdown

At the Planck scale, the smooth texture of spacetime can no longer be assumed without qualification. Distance, duration, and localization are pushed toward a regime where the background that made them mutually relatable is itself under question.

VED does not claim to provide final equations beyond this limit. It reads the Planck scale as a local *Differential Horizon* of spacetime description: the point where spacetime can no longer be treated as a smooth background texture capable of carrying arbitrary differences.

10.3 Black-Hole Singularities

Black-hole singularities should not be treated simply as objects of infinite density. Such a reading would convert a limit of description into an ordinary object of description.

From the present perspective, gravitational closure pushes the descriptive frame beyond its own ability to preserve distinguishable structure. Curvature, causal ordering, and spatial extension are driven toward a regime in which the variables of the description no longer remain mutually stable. The singularity is therefore read as a horizon-like limit of gravitational description, not as a hidden object located behind the horizon.

10.4 The Speed-of-Light Limit

The speed of light is not merely a velocity limit. It is also a boundary of temporal compression inside spacetime description. As a massive body is described in the limit approaching this speed, the required energy diverges and the temporal structure of the frame is pushed to its boundary.

From the present perspective, this is a local *Differential Horizon* of temporal description. The limit does not simply forbid a larger number for velocity. It marks the point where the temporal structure needed to describe massive motion cannot be compressed further within the same frame.

10.5 Hierarchical Interpretation

These examples do not solve quantum gravity, remove divergences by decree, or replace established physical theories. They reorganize the meaning of recurring limits. VED reads them as cases in which description approaches the edge of the differences that its own observational window can stabilize.

> Unresolved problems are not merely unrelated failures.
> Nor are they one hidden object behind physics.
> They are local and hierarchical horizon-forms of description.

The recurrence of such limits suggests a common structural pattern: whenever description approaches the disappearance, compression, or overclosure of difference, it produces a horizon-like boundary rather than a final object beyond the boundary.

The following section formulates this recurring structure explicitly as the *Differential Horizon Principle*.

11 The Differential Horizon Principle

From the preceding discussion, a structural constraint emerges. The *Differential Horizon* is not a wall located somewhere in spacetime. It is a descriptive boundary that appears whenever a spacetime-generated observer attempts to describe the condition from which spacetime itself is generated. It is the limit at which a description based on difference tries to describe the absence of difference.

> ***Differential Horizon Principle*:**
> Observation measures difference from within generated spacetime; where difference itself is no longer available, description encounters a horizon.

This principle can be expressed in three propositions. We refer to them as Propositions 1, 2, and 3 of the *Differential Horizon Principle*. Here, a generated domain means any domain in which differences have already become describable, including but not limited to spacetime.

1. Observation can measure only differences that have already entered a generated domain.
2. The absence of difference cannot appear as an ordinary state within the spacetime generated from difference.
3. Every attempt to describe the disappearance of difference from within spacetime therefore produces a horizon of description, not a final object of description.

The Chain of Necessity

These propositions are not independent.

> They form a necessary structural chain.

Because observation operates through difference, it can register only what has entered a generated domain.

Because the absence of difference would erase the condition under which such registration is possible, it cannot appear as one more state inside generated spacetime.

Because it cannot appear as an ordinary state, the attempt to describe it from within spacetime yields a descriptive boundary.

> The *Differential Horizon* is not an external prohibition.
> It is the internal signature of generated description reaching the condition that would erase its own difference.

Local and Hierarchical Horizons

The *Differential Horizon* should not be understood only as a single absolute boundary placed outside all description. The principle names a structural limit, and that limit can also appear locally. Whenever a particular observational window reaches the limit of the differences it can resolve, the same structure appears in restricted form.

Such a local *Differential Horizon* is not a second fundamental horizon. It is a local appearance of the same principle: a generated description reaches the edge of what its own differences can sustain. Section 13 gives the smallest form of this principle. There the apparent rest of the dynamic barrier is shown to be projection-dependent, and the barrier becomes a local descriptive horizon rather than an imposed resistance term.

Counterfactual Description

What lies beyond the *Differential Horizon* cannot be described as a hidden object. To describe it as an object would already place it inside a generated domain of difference. It can only be described counterfactually:

> If difference were absent, the generated description itself would no longer be available.

This formulation does not make the absence of difference into a region, substance, or state. It marks the limit at which the conditions of description fail.

Scope and Provisional Status

The *Differential Horizon Principle* does not explain existence from outside existence.

> It defines the condition under which explanation becomes possible from within a generated domain.

It does not introduce new entities. It does not propose a final unified theory of interactions. It does not describe what lies beyond the horizon.

The principle is therefore not a final metaphysical closure. It is the current formulation of a structural constraint. Its formulation may be updated as observational or mathematical windows deepen. However, as long as observation measures difference from within generated spacetime, some limit of describability remains.

> The *Differential Horizon Principle* is not a theory of what exists,
> but a principle of how existence becomes describable.

The next section positions VED within this framework.

12 The Position of ved

The preceding section formulated the *Differential Horizon Principle* as a limit of describability from within generated spacetime. Within that framework, the position of VED (*Vortical Enclosure Dynamics*) can be stated more precisely.

> VED does not describe the outside of the *Differential Horizon*.

The *Differential Horizon* is not a literal spatial boundary, and what lies beyond it cannot be treated as a hidden object. VED therefore does not convert the *ungenerated* into an object, region, or state. It describes why such conversion fails. When a generated description attempts to reach the condition from which its own difference is generated, it encounters a descriptive boundary.

Generative Order

VED proposes a generative order:

$$\text{difference} \longrightarrow \text{gradient} \longrightarrow \text{motion} \longrightarrow \text{spacetime.} \tag{1}$$

This order is not merely explanatory. It is a physical constraint on what can appear inside generated description. If spacetime is generated from difference, then the absence of difference cannot appear as an attainable state within spacetime. To appear as a state, it would already have to be differentiated from other states.

> The absence of difference cannot be made into one more difference.

This is why the *Differential Horizon* appears. It is the descriptive boundary produced when a spacetime-generated description attempts to describe the condition that would remove the difference on which that description depends.

Open Generative Framework

VED is not a final closure of physics. It does not explain existence from outside existence, nor does it replace quantum field theory, relativity, or other successful descriptions inside their domains. It is an open generative framework for describing why complete closure cannot be achieved within generated description.

> VED does not close theory.
> It describes why closure fails from within theory.

This failure is not simply a gap in current knowledge. It is the structural consequence of describing from within a domain whose own conditions depend on difference. As observational and mathematical windows deepen, the formulation of VED may change. But as long as physical description measures difference from within generated spacetime, the disappearance of difference cannot be reached as an ordinary object of description.

Bridge to the Dynamic Barrier

Section 13 gives the smallest form of this position in a finite dynamical model. There, complete closure is not blocked by an external wall. It is blocked by the fact that a spacetime-generated model cannot erase the difference that makes its own description possible.

The dynamic barrier therefore does not add a new metaphysical object. It localizes the role of the *Differential Horizon* inside a minimal system of closure, rotation, and dissipation. What appears as a resistance term is reinterpreted as a local descriptive horizon: the model reaches the limit of the differences by which it can describe its own closure.

> What has been reached is not an answer,
> but the structure of a boundary.

13 The Dynamic Barrier as a Local Differential Horizon

13.1 From Phenomenological Barrier to Dynamic Feedback

The earlier use of a κ_B barrier term should be read as a phenomenological approximation. It records that complete closure is resisted, but it does not explain why the resistance exists. In schematic form the previous description was

$$B(L) = -\kappa_B \ln\left(1 - \frac{L}{L_{\max}}\right), \tag{2}$$

so that the resistance diverges as L approaches $L_{\max}$.

This section replaces that interpretation. The barrier is not introduced as an external prohibition. It is generated by the same motion that attempts to close the system: closure increases the velocity of closure, closure velocity generates rotational feedback, and rotational feedback bends the trajectory away from complete closure.

The resulting model is a local spacetime-side realization of Proposition 1 of the *Differential Horizon Principle*. Complete closure, $L = 1$, corresponds not to an ordinary state inside the model, but to the boundary at which no internal difference remains available to sustain the generated description—the local signature of the *ungenerated* domain. The model does not allow this limit to be reached from outside by imposing a wall. It shows how the approach to that limit becomes dynamically unstable from within the coupled motion of closure, rotation, and dissipation.

> Complete closure is not prohibited by a wall.
> Rather, motion toward complete closure destabilizes its own existence.

Let q be an unconstrained closure coordinate and define

$$L = \frac{1}{1 + \exp(-q)}. \tag{3}$$

The dynamics are

$$\frac{dq}{dt} = v, \tag{4}$$

$$\frac{dv}{dt} = AL - B\Omega^2 L - Dv, \tag{5}$$

and

$$\frac{d\Omega}{dt} = CLv - E\Omega. \tag{6}$$

Here v is closure velocity and Ω is the rotational or vortical feedback generated by closure motion. There is no explicit κ_B potential term. If a barrier appears, it must appear as a dynamical effect of the coupling among L, v, and Ω.

13.2 Fixed Points and the Degenerate Manifold

The near-closure balance is easiest to see in the conserved-rotation limit $E = 0$. For $L \simeq 1$ and $B > 0$, the closure acceleration is approximately

$$\frac{dv}{dt} \simeq A - B\Omega^2 - Dv. \tag{7}$$

The closure drive is balanced when

$$\Omega \simeq \sqrt{\frac{A}{B}}, \qquad v \simeq 0. \tag{8}$$

More exactly, in the full three-variable system with $E = 0$, the fixed-point condition gives $v^* = 0$, while Ω^* is not forced to vanish by rotational dissipation. The remaining condition is

$$L^* \left(A - B(\Omega^*)^2\right) = 0. \tag{9}$$

For $L^* > 0$, this yields

$$\Omega^* = \sqrt{\frac{A}{B}}, \tag{10}$$

while L^* itself remains undetermined. Thus the conserved-rotation limit contains a one-dimensional fixed-point manifold

$$\left\{ \left(L, 0, \sqrt{\frac{A}{B}} \right) : L \in (0, 1) \right\}. \tag{11}$$

This degeneracy is lifted by any $E > 0$, which selects the asymptotic near-closure branch instead of leaving L neutral.

In this projection the system appears to approach rest. However, this apparent rest does not mean that the system has reached complete closure. It means that the observable closure velocity has been balanced by rotational self-resistance.

This distinction matters. The sigmoid map guarantees $0 < L < 1$ for finite q, while the formal limit $L = 1$ requires $q \to \infty$. The degenerate endpoint is therefore not an ordinary point in the finite state space. It is the boundary at which the generated description would lose the difference needed to describe its own generation. The dynamic barrier is the local trace of this boundary inside the finite dynamics.

13.3 The Collapse of the Critical Dissipation Interpretation

Previous scans identified a numerical threshold

$$D_c \approx 0.0875 \tag{12}$$

[1] in the case $E = 0$. That value remains diagnostically useful, but its interpretation must be weakened. It should not be read as a true bifurcation or as a universal dynamical critical point. It is a finite-window relaxation threshold: above this value the selected observation window sees the trajectory relax rapidly enough to appear quasi-closed, whereas below it the same window continues to resolve persistent phase rotation.

This reinterpretation is not a retreat from the barrier argument. It clarifies it. The question is not whether a single scalar threshold separates structure from non-structure absolutely. The question is whether a spacetime-generated model can reach the absence of difference from within its own finite coordinates. The observed answer is negative. Changing the window may change the apparent relaxation threshold, but it does not turn the formal closure endpoint into an attainable state.

13.4 Additional Observation Windows

The following diagnostics use the same ODE system. They are observation windows, not new physical variables. Their purpose is to test whether apparent convergence in one projection is equivalent to genuine disappearance of motion. In other words, convergence and dispersal can produce the same appearance when the observation window is too narrow to distinguish them. The point is methodological: if a claim of rest disappears when the same trajectory is viewed through a richer reconstruction, then the rest was not an absolute state but a projection-dependent description.

Figure 4 plots the conserved rotation case in the centered plane $(v, \Omega - \sqrt{\frac{A}{B}})$. Even when the amplitude decays, the trajectory retains phase rotation around the closure-balance point. Thus apparent convergence depends on the observation window: a scalar closure plot can look nearly static while the phase projection still displays rotation.

The sliding-window extent in Fig. 5 shows approximately exponential relaxation over finite windows. This supports the reinterpretation of D_c as a finite-window relaxation threshold rather than a sharp dynamical critical point.

[1] The numerical value depends on the empirical instability threshold used to define relaxation in a finite observation window. For the criterion $I(D) < \theta$, repeating the analysis on the same integration data with $\theta \in \{0.01, 0.02, 0.03, 0.05, 0.07, 0.10\}$ gives $D_c(\theta) \in [0.075, 0.125]$. This dependence is not a defect of the interpretation; it is evidence that D_c is a finite-window relaxation threshold, not a sharp bifurcation point.

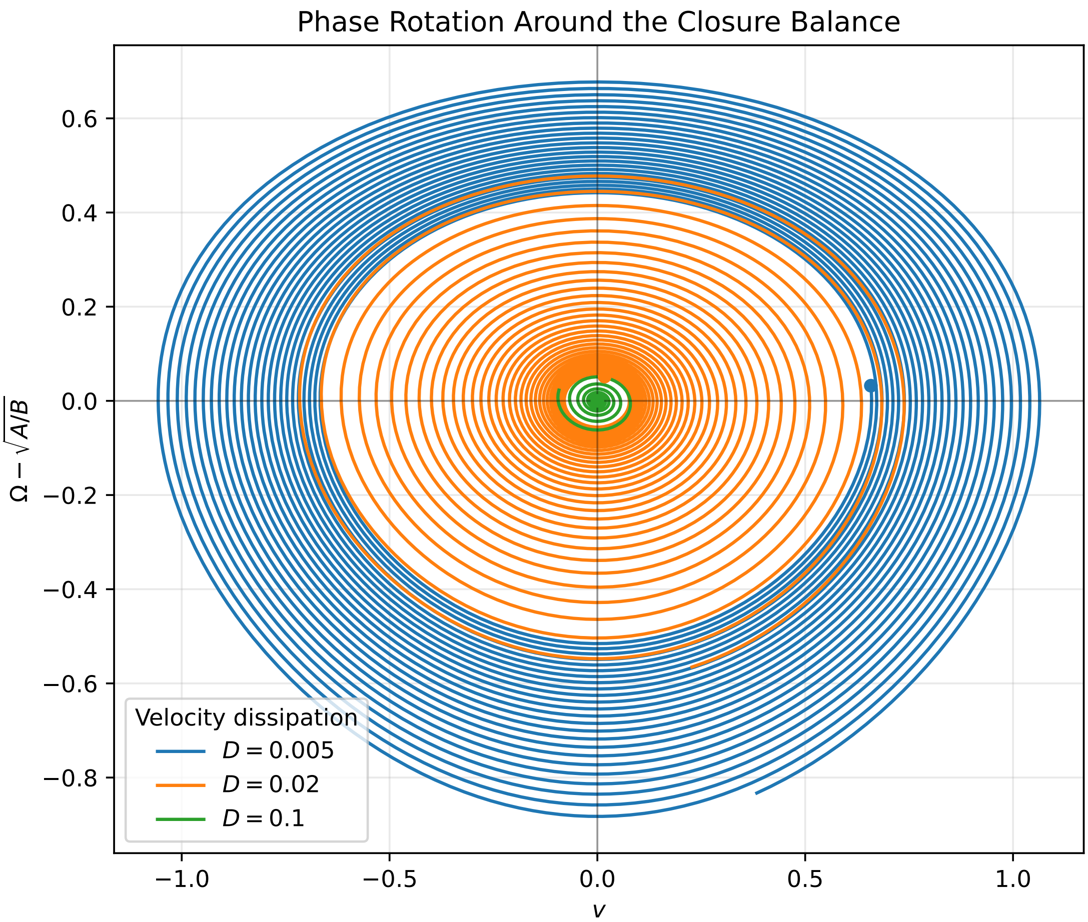

Figure 4: Phase rotation around the near-closure balance in the conserved rotational-energy limit. The radius decays under dissipation, but phase rotation remains visible in the centered projection.

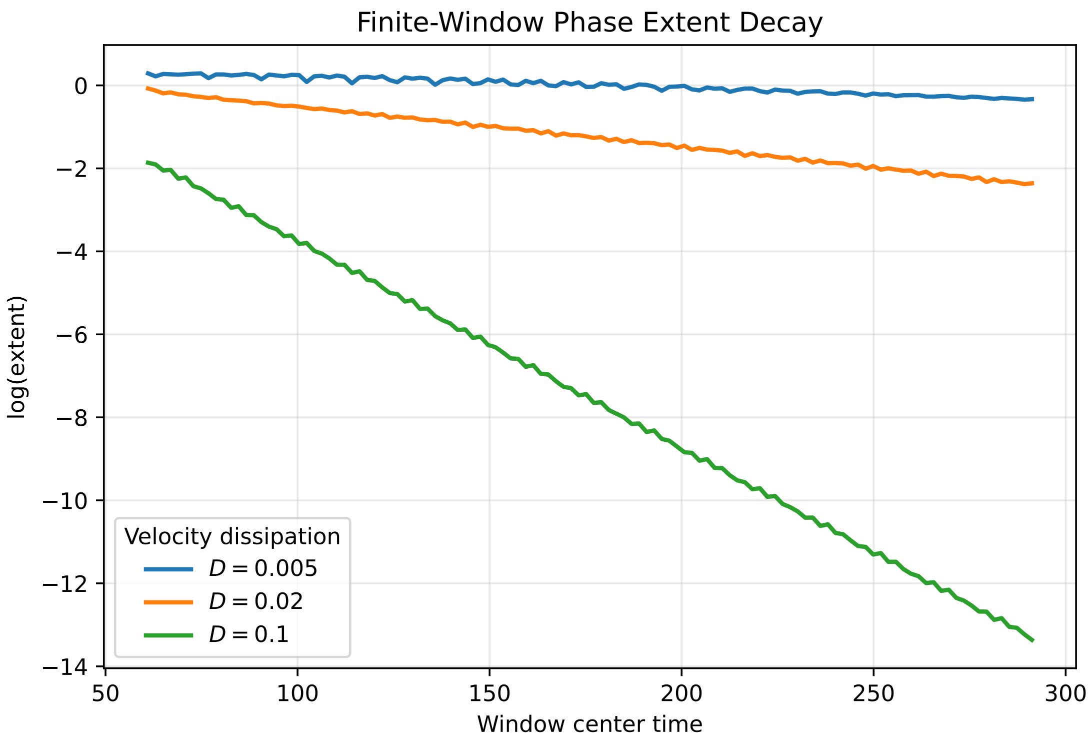

Figure 5: Finite-window decay of phase-space extent under velocity dissipation. The approximately linear trend in log scale supports the relaxation-threshold interpretation.

The spectrum of $v(t)$ in Fig. 6 shows a fundamental frequency together with harmonic structure: the dominant peaks appear at harmonically related frequencies rather than as a featureless decay of spectral power. These peaks are generated by the nonlinear sigmoid closure map and the quadratic feedback term $B\Omega^2 L$. They show that the near-closure state is not reducible to a single static scalar value.

Figures 7 and 8 make the same point in reconstructed state spaces. The delay embedding of $v(t)$ and the higher-derivative projection show that what appears as rest in one window reappears as motion in another. The trajectory is therefore not exhausted by the scalar statement "v becomes small."

Finally, Fig. 9 examines the residual spectrum of the detrended radial envelope

$$r(t) = \sqrt{v(t)^2 + \left(\Omega(t) - \sqrt{\frac{A}{B}}\right)^2}. \tag{13}$$

The envelope itself contains modulation. Its residual spectrum again displays dominant peaks at harmonically related frequencies, now in the decay envelope rather than directly in $v(t)$. This confirms that the apparent barrier is not a static potential wall; it is a delayed rotational self-resistance whose signatures depend on how the motion is observed.

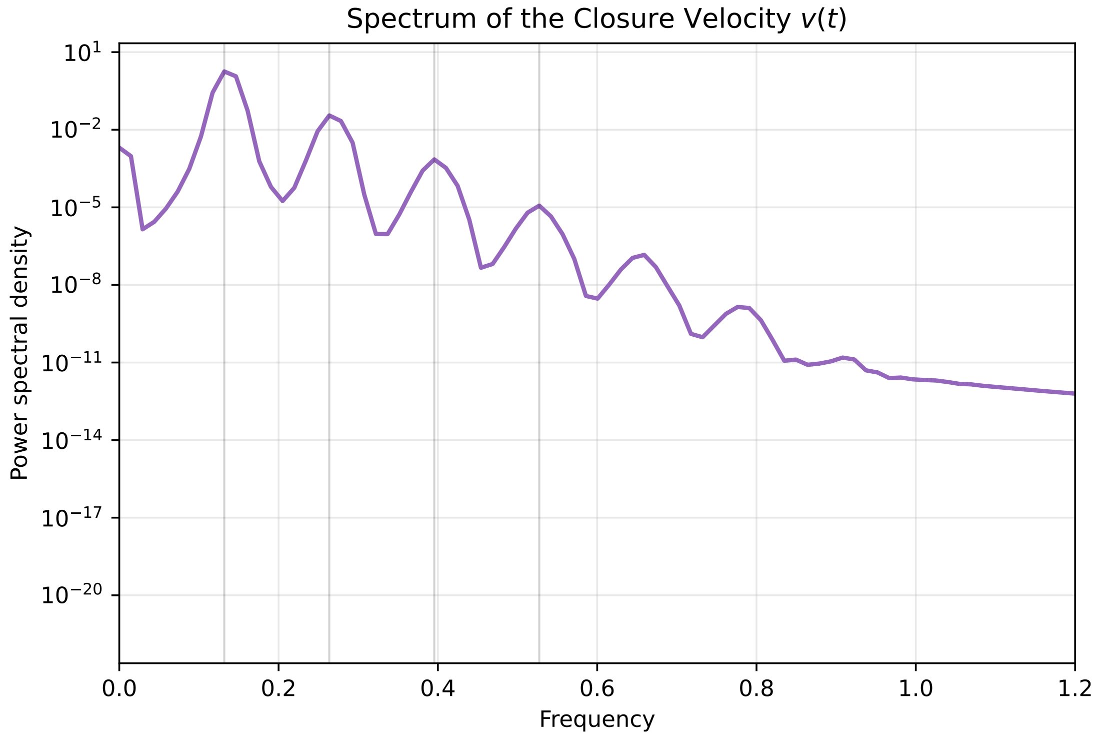

Figure 6: Spectral content of closure velocity generated by nonlinear feedback. Dominant peaks occur at harmonically related frequencies.

13.5 Apparent Rest as a Projection

The phrase "quasi-closure" should therefore be understood carefully. It does not mean that the system has reached a final static object. It means that, in a selected finite observation window, the closure coordinate, closure velocity, rotational feedback, and dissipative losses have entered a coherent balance.

This balance is projection-dependent. In the $L(t)$ projection, the system may appear nearly closed. In the $(v, \Omega - \sqrt{\frac{A}{B}})$ projection, it may still rotate. In a delay embedding, the same signal can retain a loop-like structure. In derivative space, small motion can reappear as higher-order curvature.

This is not only a numerical artifact of plotting. At finite q the sigmoid map is analytic, so a local expansion of $L(q)$ contains higher powers of the closure coordinate. Together with the quadratic feedback term $B\Omega^2 L$, these nonlinearities mix frequencies and generate harmonic content. The apparent rest of the scalar closure coordinate can therefore coexist with residual motion in richer reconstructions.

The observation windows form a hierarchy. Each window suppresses some components of the trajectory while exposing others. Adding a phase portrait, a spectrum, a delay embedding, or a derivative-space reconstruction does not add new physics; it asks whether the same finite trajectory still contains difference after a narrower projection has declared it to be at rest.

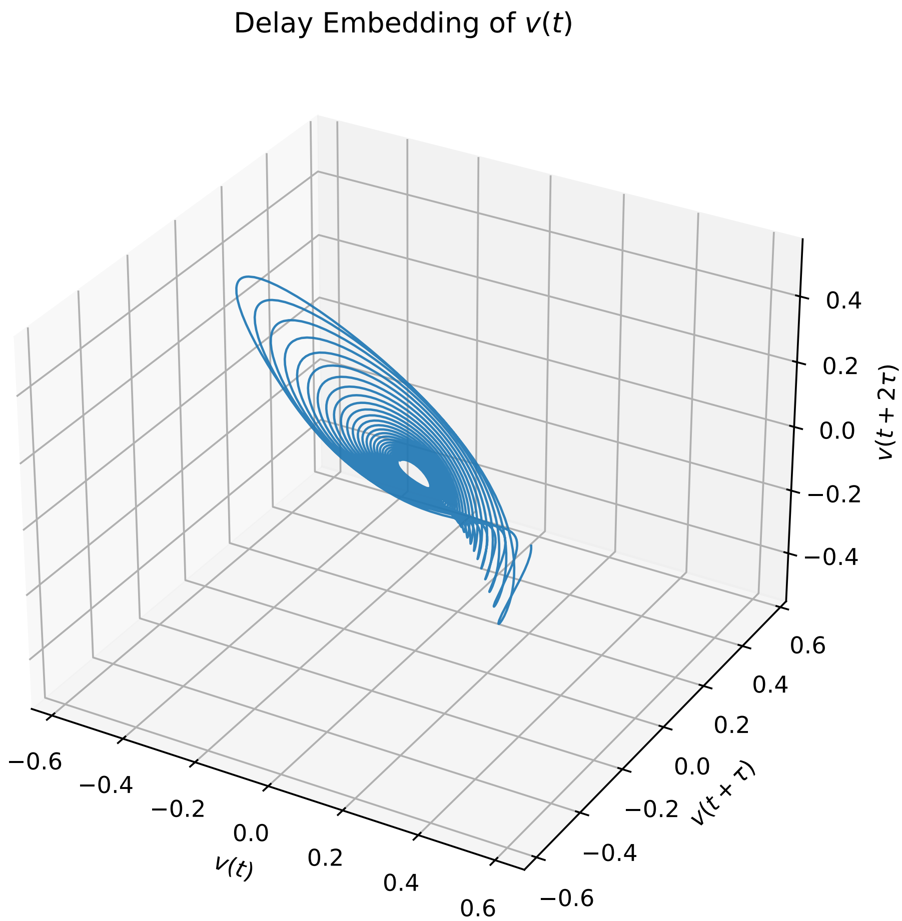

Figure 7: Delay embedding of closure velocity as an observation-window reconstruction. The reconstructed loop shows motion hidden by scalar convergence.

Higher-Order Phase Structure

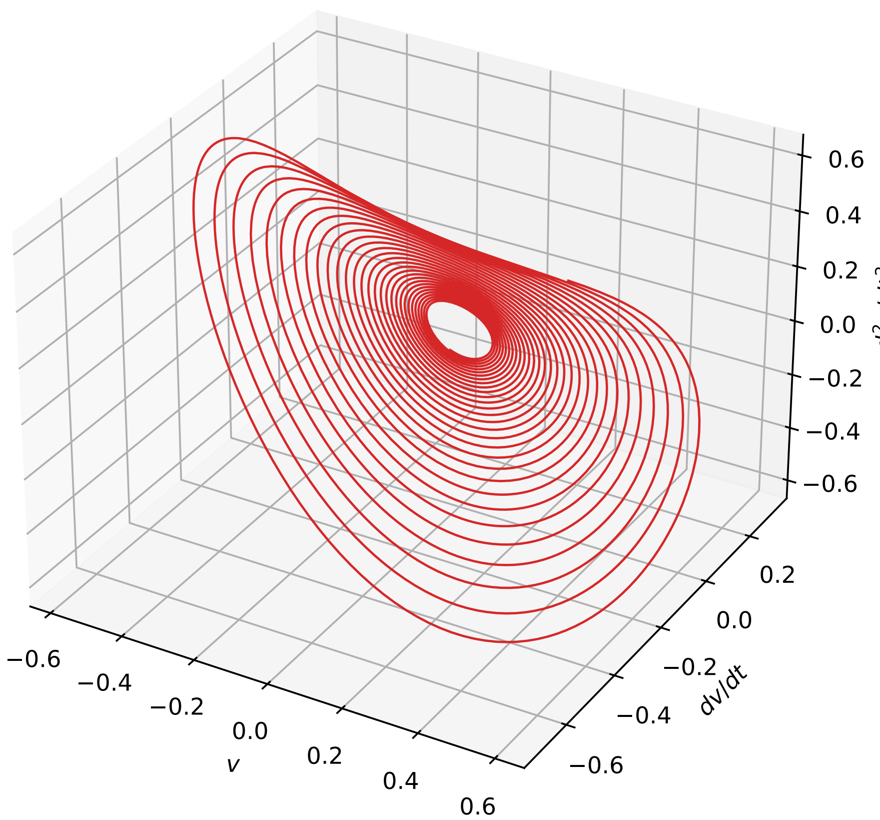

Figure 8: Higher-order phase structure in closure-velocity derivatives. Small projected motion reappears as curvature in derivative space.

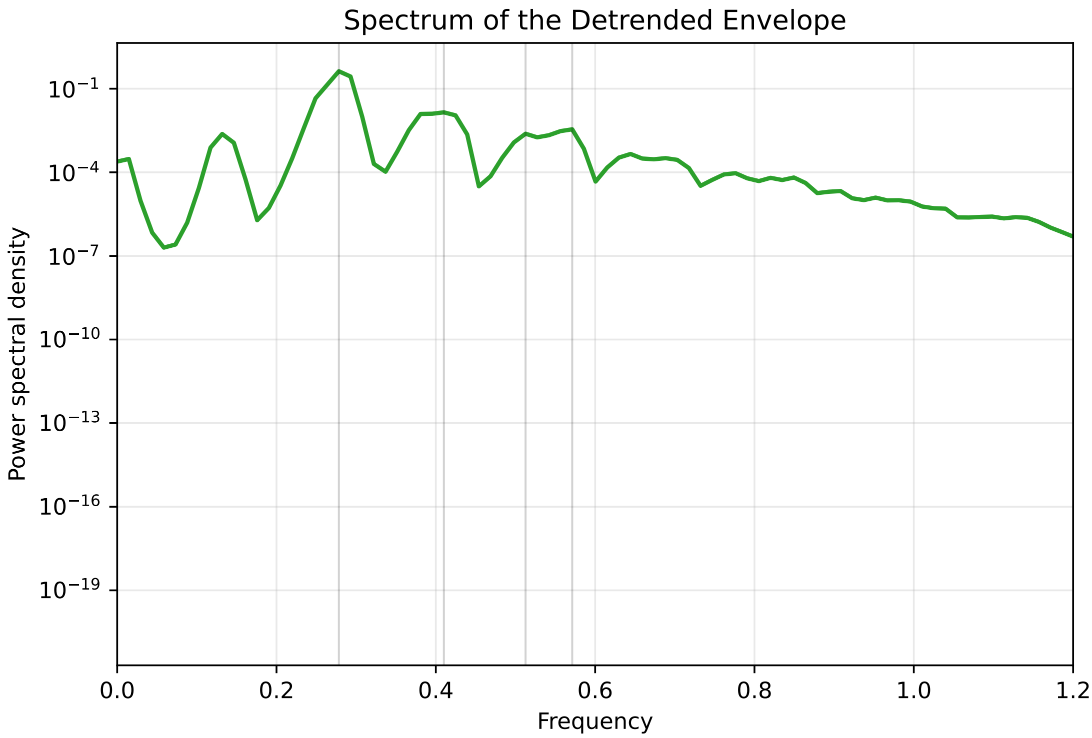

Figure 9: Residual spectrum of the detrended phase-envelope radius. Envelope modulation remains after the dominant decay is removed.

The dynamic barrier is therefore not a hidden external wall. It is the condition under which closure can remain describable without erasing the difference required for description. Dissipation does not destroy structure. It determines whether the remaining motion can be organized as quasi-closure rather than persisting as unresolved rotation.

13.6 The Dynamic Barrier as a Local Differential Horizon

The revised interpretation is the following. The dynamic barrier is not merely a resistance term preventing complete closure. It is the internal signature of a descriptive boundary: a spacetime-generated model cannot reach the absence of difference from which spacetime itself is generated.

In this sense, the barrier is a local *Differential Horizon*. It is not the global statement of the *Differential Horizon Principle* itself, but a finite dynamical model of how Proposition 1 appears from within a generated description. The model does not prove the *Differential Horizon Principle* from mechanics. Rather, it shows the spacetime-side form of the same limitation: the attempt to remove all difference generates dynamics that prevent the removal from being completed inside the model.

The previous κ_B term can therefore be retained only as an effective phenomenological shorthand. In a stable quasi-closure regime, and only in a projected or coarse-grained sense, one may write

$$\kappa_B^{\mathrm{eff}} \sim \Omega^2, \tag{14}$$

but this is not a fundamental potential. It is a projected measure of delayed rotational self-resistance.

> VED does not impose non-closure as an external law.
> It derives non-closure as the only dynamically coherent regime capable of sustaining structure.

Complete closure would remove the difference by which the model describes its own generation. For that reason it is not reached as an ordinary dynamical endpoint. It appears instead as a local Differential Horizon: the boundary at which a spacetime-generated description can no longer complete its own closure from within. This makes Section 13 not a peripheral application of the broader framework, but its smallest form: the structure of the *Differential Horizon Principle*, made local enough to be solved as a finite ODE.

14 Conclusion

This volume has traced the structure of divergence, renormalization, and the limits of modern physical theory.

It has introduced the *Differential Horizon* as a descriptive boundary: the limit at which a description based on registered difference fails to convert the absence of difference into an ordinary observable state.

> It does not reveal a hidden object.
> It clarifies where object-description ceases.

The *un-generated* cannot be described as a region, state, or object inside generated spacetime.

It appears only as the failure of conversion: the point at which absence of difference cannot be made into one more difference.

> The boundary is not an object,
> but a condition of description.

Divergence appears when a descriptive window is pushed toward the limit of the differences it can represent as finite quantities.

Renormalization does not erase that limit. It re-expresses boundary contact as finite relations inside a chosen window.

These are not separate phenomena.

They are structural responses
to limits of describability.

The unresolved problems considered in this volume should therefore not be read as independent failures, nor as different signs of one hidden object. They are local or hierarchical horizon-forms: places where successful descriptions reach the edge of the differences their own variables can sustain.

VED does not close physics from above. It provides an open generative framework for understanding why complete closure cannot be achieved within generated description.

Several questions remain open.

The precise mathematical structure of the *Differential Horizon*— its local forms, hierarchical relations, and connection to known geometric structures— awaits further development.

The connection between this framework and existing approaches to quantum gravity, holography, and information-theoretic methods remains to be made explicit.

The dynamic non-closure derived in Section 13 should be read in this light. It is not a separate phenomenological claim, but the dynamical realization of Proposition 1 of the *Differential Horizon Principle*. What the principle states as a structural impossibility— that the *un-generated* domain cannot be reached— the model exhibits as a coherence condition: structure persists only in the regime where closure remains incomplete. Non-closure is therefore not an additional rule, but the only dynamical regime in which generation itself can be sustained.

The *Differential Horizon* is not a place that can be reached.

It marks the limit of what can be described.

Nothing further can be said from within the same descriptive frame.

Differential Horizon Map

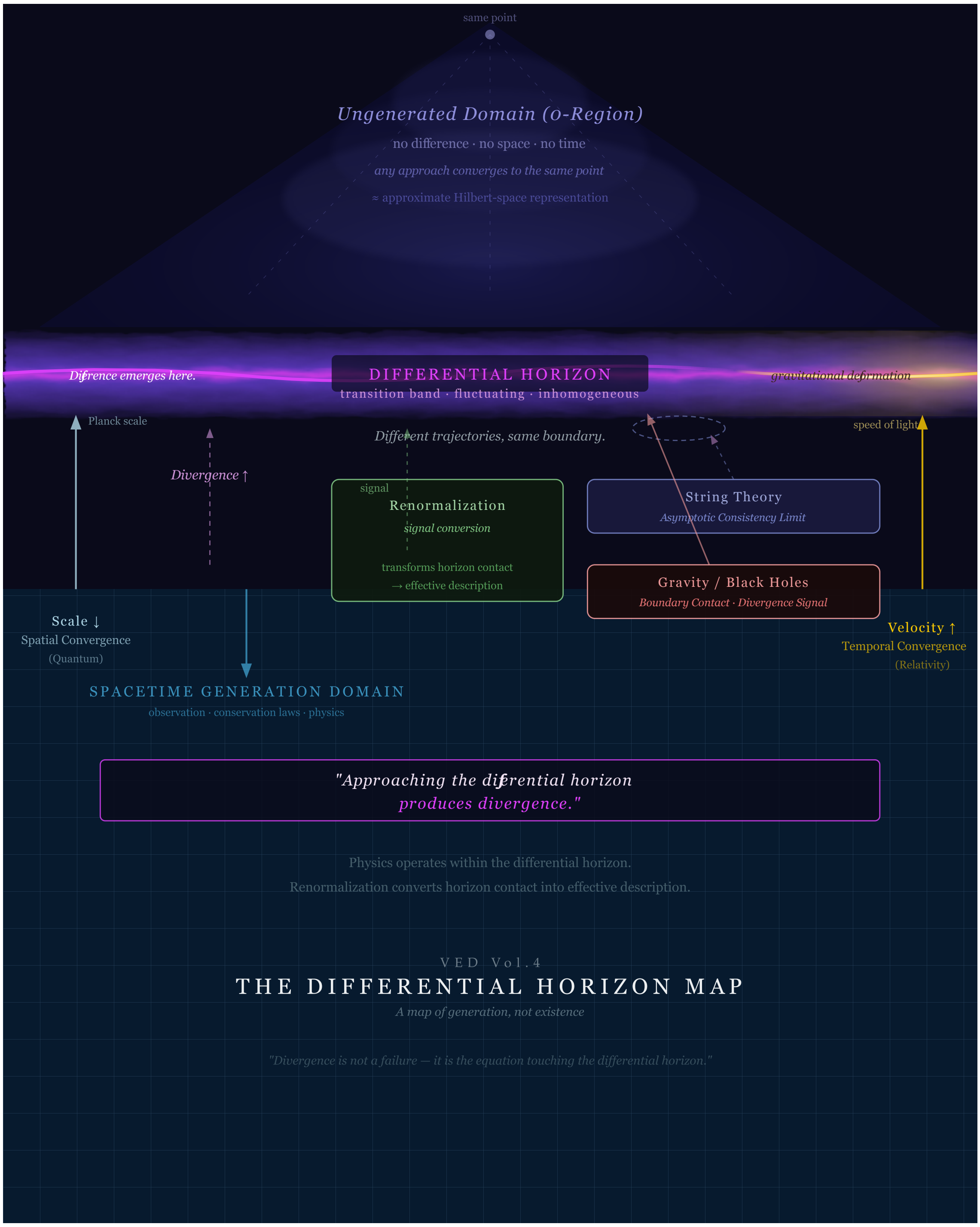

Figure 10: The Differential Horizon map. This schematic summarizes the structure developed in Vol. 4. The *un-generated* limit is not directly described as an object. The *Differential Horizon* marks the descriptive boundary at which divergence and boundary contact appear. The spacetime generation domain is the region in which physical theories operate on registered differences. Renormalization converts boundary contact into effective description within a finite observational window. Quantum theory, relativity, string theory, gravity, and black-hole physics expose related local horizon-forms through different trajectories. Section 13 gives the smallest form of this structure as a finite ODE.